RP9 — Error log of Revision_Deflation

Provenance of PROJECT spacetimes, Revision_Deflation: every error made during the work on the revision, how it was found, and how it was corrected

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November 13, 2026

Purpose

Hard Rule 7 of the task specification forbids shortcuts and demands that problems be explained. As the provenance P9 does for the main project, this file records every error that was made while the revision was carried out — including errors that were corrected before anything was committed. For each it states what was wrong, how it was detected, what was done, and where the corrected statement can be checked. An error that is not listed here and is found later is to be added, not silently repaired.

The errors are numbered R-E1, R-E2, ... in the order in which they were recorded.

1. Errors of physics and of mathematics

No.	What was wrong	How it was found	Correction	Where to check
R-E1	For unequal rates (7+1) it was first asserted that the coordinate that becomes energetic first decides the number of created particles whenever the two crossings are at least three e-foldings apart. This is false when the second direction deflates more slowly.	The check failed for a slower second direction.	The criterion is that the second physical wavenumber is still below $e^{-3}E_{\perp}$ at the first crossing.	check `71d-M3`
R-E2	It was first asserted that all eight bounded solutions of the 4+4 field fall by the same factor. This holds only when one rate governs.	The check failed for three unequal rates.	With unequal rates the factor depends on the internal state (spread 0.40 at $S_{\mathrm{WKB}} = 30$); stated so.	check `44d-M3`
R-E5	The critical value of C at which matter becomes abnormal was required to agree with the estimate $\pi^2 m^2/4k_F^2$ of the main project within 60 per cent at all densities.	The check failed at low density.	The estimate is reached only at high density; at $k_F = 0.2\,m$ the critical value is 2.46 times larger.	check `def-H5`; RS4, Sec. 3.1
R-E6	The effect of the loss of mass of bound pairs on the Hubble constant was first asserted as “at most 2 km/s/Mpc” — a guess — and computed with $\alpha_{VS} = 3$, where the formula for the binding energy is not valid.	Comparison of the assertion with the computed numbers; review of the range of validity.	Examples with $\alpha_{VS} \le 1$; at constant G_N the gain is at most $+0.48$ km/s/Mpc.	check `def-C4`; RS3, Sec. 4

No.	What was wrong	How it was found	Correction	Where to check
R-E7	A bound on the rate was derived from the constancy of m_p/m_e by assuming that the electron mass follows the four-dimensional vacuum value of the Vorton-Scalar. Under that assumption the mass is the product of a coupling that grows and a vacuum value that falls, and is constant.	Re-derivation of the reduction of the couplings.	The row was replaced by the bound from the Fermi constant at fixed masses.	check `def-C5`; RS3, Sec. 3
R-E9	It was asserted that the amplitude σ_8 of clustering falls when Newton's constant grows with the deflation.	The computed number has the other sign when the Hubble constant is the one that keeps the sound-horizon angle.	Both cases are stated: -0.317 at fixed Hubble constant, $+0.102$ at fixed angle. (These two numbers were themselves superseded: see R-E33.)	check `def-D4`; RS3, Sec. 5
R-E11	The coherence length of the Ginzburg–Landau energy was computed with a coefficient too large by the number of internal states, a factor 2 in the length.	Comparison with the weak-coupling value $0.1333\, v_F/T_c$.	Corrected; the weak-coupling formulas are stated to hold only for $T_c < 0.05\, E_F$.	check `def-P3`; RS4, Sec. 2.3
R-E12	The law "the condensation energy has twice the logarithmic slope of the gap" was asserted for all moments; and a counter of strongly coupled moments mixed the two particles.	The check failed at the strongly coupled moments.	The law is stated for weak coupling; the counter is per particle.	check `def-P1`
R-E13	The text of a check said that the fermion mass in self-bound matter "falls to" zero along the clock. It is zero at every moment after the threshold.	Reading the data file.	Text corrected.	check `def-P7`
R-E14	A routine of the main project that finds the melting temperature treats "the Hartree potential is not defined at 1.05 times the vacuum value" as "melted". For $m_s/m = 0.3$ and $C \geq 32$ this produced temperatures that are artefacts; a check of monotonic decrease had passed with them.	A jump in the figure.	The program marks such moments as not defined (26 of 30 are defined). The table of the main project, for $C \leq 9$, is not affected.	check `def-H7`; RS4, Sec. 4
R-E17	The table of exponents listed G_{VS} as g_{VS}^2/m_s^2; the main registry defines it as $1/(\sqrt{2}\, v_{VS}^2)$ (the exponent is the same).	Comparison with the main registry.	Corrected.	`data/coupling_exponents.csv`
R-E18	The independent Python script had the wrong sign of one term in the three special Killing vectors of the case of equal rates. The Rust program was right.	The independent check failed.	Sign corrected, with the derivation in a comment.	check `I44d-G4`, `I71d-G4`
R-E21	The first table of the growth of the gap multiplied the logarithmic slope by the pairing strength of the radial mode alone and found 0.96 and 0.77 instead of 1.	Disagreement with check `def-H3`, which uses the sum of the strengths of both modes.	The table uses the pairing strength of the vector channel, $\lambda_S + \lambda_P$; the check says so explicitly.	RS1, Sec. 3.2

No.	What was wrong	How it was found	Correction	Where to check
R-E23	**The label H2 was attached to the wrong hypothesis.** In the main project H2 is the author's hypothesis that the superfluid state is the Higgs field. In the programs and first documents of the revision the label was used for a different assumption — that the vacuum value of the Vorton-Scalar sets the electroweak scale. The actual consequence of H2 on a deflating spacetime had therefore not been examined.	Reading the supplement S2 of the main project while writing RS2.	The other assumption is now called HV. The consequence of H2 was derived and computed: the electroweak scale follows α_{VS}, and atomic clocks bound the rate to $1.16 \times 10^{-6} H_0 / n_{\mathrm{EW}}$.	check `def-C6`; RS2, Sec. 5; RS3, Secs. 1, 3
R-E25	The statement of a check, and with it the supplement RS1 and the table of RP0, spoke of "90 pair channels". The program had counted each channel once for each of the three momenta: there are 15 representative channels on each spacetime, 30 in all, evaluated at three momenta and 13 angles (1170 numbers). The comparison itself was complete and correct.	Comparison with the data file of the main project while writing the manuscripts.	The program counts the distinct channels; the statement and the documents say 15 on each spacetime.	check `def-K1`; RS1, Sec. 3.1
R-E26	The locally measured Hubble constant was written as "73" in two statements of the program, in a figure and in RS3, and the distance of the model from it as 5.6 km/s/Mpc. The main project compares with 73.50 ± 0.81 km/s/Mpc; the distance from 67.36 is 6.1. No conclusion changes (the gain of at most $+0.48$ remains below a tenth of it).	Comparison with the manuscripts of the main project while writing the manuscripts of the revision.	The program takes the value 73.50 and its uncertainty as stated; statements, figure and RS3 corrected.	checks `def-C3`, `def-C4`; `figures/fig_hubble`
R-E28	The summary of RS3 gave the temperature of the quanta that the deflation creates as "10^{-33} K at the allowed rate". That is its value at the bound of lunar laser ranging (2.9×10^{-33} K); at the largest rate that any of the three bounds allows it is 2.1×10^{-31} K. The conclusion (nothing is created today) is not affected.	The check of the numbers of the running text (`check_numbers.py`), written for the manuscripts, found that the number was stated by no check.	The text gives both values; check `def-D6` states them.	RS3, Sec. 0 and 7; check `def-D6`
R-E30	The caption of the table of states of the order parameter in the drafts of the manuscripts gave the unit of the two gaps of a state as "the square of the magnitude of the order parameter". For the half-paired state the tabulated value 1 would then have to be 2. The numbers are in units of Δ_F^2, the square of the gap of a unitary state, and each state is taken at the magnitude at which its potential is stationary; the data file says so, and RS1 gave no unit. The numbers themselves were right.	Re-reading the pages of the finished manuscript against the identity for the gap matrix.	Caption corrected; RS1 states the units.	manuscripts, Sec. X; RS1, Sec. 5.2; `data/order_parameter_states.csv`

No.	What was wrong	How it was found	Correction	Where to check
R-E33	**The growth equation of RS3 was not the limit of the eight-dimensional theory.** RS3, Sec. 5, the manuscripts and the check `def-D4` integrated the familiar equation $d^2\delta_m/dx4^2 + 2H\,d\delta_m/dx4 = 4\pi G_N(x4)\rho_m\delta_m$ and called it the Newtonian limit of the theory with assumptions A1 and A2. It is not. With unperturbed extra scale factors the eight-dimensional equations give a friction reduced by $\mathrm{gsum}/2$ and, depending on what is assumed about the agent that keeps the rates constant, a reduced source as well. The quoted results (-0.317 and $+0.102$ per unit of gsum/H_0 for the amplitude; factors 1.0027 and 1.0589 for the growth rate) belong to neither case.	The derivation of the perturbation equations for RS5 by computer algebra; then the complete Einstein–Boltzmann calculation, whose growth disagreed with that equation by a quarter of gsum/H_0.	Two closures are computed and a third analysed; RS3, Sec. 5, the manuscripts and the check state the results of each and that the growth is not a prediction of the hypotheses. The routine keeps the four-dimensional equation for comparison only.	checks `def-D4`, `def-E8`, `Idef-P4` to `Idef-P7`; RS5, Sec. 4; RS3, Sec. 5
R-E34	**The fall of the Hubble constant was underestimated by six per cent.** The command `cosmology` held the redshift of last scattering and the helium fraction at their values without deflation when it kept the angle of the sound horizon. With weaker gravity in the past, last scattering is later and the helium fraction lower, and both enlarge the sound horizon further. The slope was given as -0.911 instead of -0.969, the values for $\mathrm{gsum} = 0.01, 0.03, 0.1\,H_0$ as 66.74, 65.48, 60.77 instead of 66.70, 65.36, 60.37 km/s/Mpc, and the changes at the bounds of lunar laser ranging and nucleosynthesis as -0.199 and -4.13 instead of -0.211 and -4.38 km/s/Mpc. No conclusion changes; the fall is larger.	The copy of the recombination program, run on the deflating background for RS5, moved last scattering.	`cosmology` computes the full thermal history and tabulates the old approximation beside it; an independent recombination code confirms both (`Idef-C2`). The examples with a varying rest mass (RS3, Sec. 4) still use the approximation, and say so.	checks `def-C2`, `def-C3`, `def-B2`, `Idef-C1`, `Idef-C2`; RS3, Sec. 2; RS5, Sec. 6
R-E35	The earlier version of RP0, Sec. 5, and of RS3, Sec. 9, named as the simplest assumption for the perturbations that "the scale factors of $x5, x6, x7$ are not perturbed, and the perturbations of ordinary spacetime obey the four-dimensional equations with the Newton's constant of the moment". The second half does not follow from the first and is not a consistent system: the eight-dimensional equations contain $H_e = H_{\mathrm{cf}} - a_{\mathrm{sp}}\mathrm{gsum}/2$ in place of the conformal Hubble rate, and of the four scalar Einstein equations only two can hold with matter alone.	The derivation by computer algebra; and in the first copy of the Einstein–Boltzmann program the equation of the shear, tested numerically along a solution, failed at the first order in the rate.	Assumption A3 with its closures (RS5, Secs. 1 to 3); every result is marked as depending on the closure or not.	checks `I44d-P1`, `I71d-P1`, `Idef-P3`, `def-E4`, `def-E5`; RP0, Sec. 5

No.	What was wrong	How it was found	Correction	Where to check
R-E38	A unit test of the growth equations was first written with guessed first-order coefficients of the growth rate ($-1/10$, $-1/2$, $-3/5$ times gsum$/H$). The test failed. The coefficients, derived by a series solution, are $-9/16$, $-13/16$ and $-11/16$; and the test model needed a vanishing cosmological term, which otherwise shifts the numbers by 0.006.	The unit test.	Derived, confirmed by computer algebra, and tested to 10^{-4}.	unit test `growth_with_deflation_follows_the_first_order_laws`; check `Idef-P4`
R-E43	The response of the helium fraction to the smaller Newton's constant at nucleosynthesis was computed as $\frac{43}{7}(e^s-1)$ equivalent neutrino species for every model. That is right for the standard radiation content only. In a model with extra radiation of the density ΔN during nucleosynthesis the expansion rate is proportional to $e^s(\frac{43}{7}+\Delta N)$, and the equivalent number is $(\frac{43}{7}+\Delta N)(e^s-1)$. The difference is of the second order (the product of the rate and of ΔN): 4×10^{-5} in the helium fraction for $\Delta N=1$ at the rate that lunar laser ranging allows, and 4×10^{-4} for $\Delta N=0.5$ at the rate that nucleosynthesis allows, where it changed the fourth decimal of two entries of a table.	Re-reading the routine while the first run of the command `families` was under way.	The routine uses the exact correspondence; a unit test checks it; the commands `background` and `families` were run again. Models without extra radiation are not affected, to the last bit.	`eb_cosmo.rs`; unit test `newtons_constant_at_nucleosynthesis_lowers_the_helium_fraction`; RS5, Sec. 5.1

2. Errors of numerical method

No.	What was wrong	How it was found	Correction	Where to check
R-E3	The test of the order of convergence of the integrator compared the result with the exact value at a *finite* end point, where the two differ by the residual of the projection, not by the error of integration.	The apparent order was wrong.	The test compares runs with different steps.	unit test `the_integrator_converges_with_order_four`
R-E8	The number of created scalar quanta was read off where the physical wavenumber was 300 times the frequency; the residual, 6×10^{-4}, was large compared with the smallest numbers.	The deviations of the check.	Read off at 3000 times: 1.9×10^{-5}.	check `def-B3`
R-E10	The "epoch of strong self-interaction" was first defined by the threshold 1 cm^2/g, which the example particle exceeds at all times.	The interval came out as the whole range.	Threshold 100 cm^2/g.	check `def-D3`

No.	What was wrong	How it was found	Correction	Where to check
R-E36	The closure “Shear” was first built into the copy of the Einstein–Boltzmann program, with the equation of the shear integrated as an equation of motion. In that form the equations are stiff: the momentum density of the agent relaxes at the rate $k^2/3H_{\mathrm{cf}}$. The explicit integrator of the main project could not follow it; even at zero rate the temperature spectrum differed from that of the main project by 1.6×10^{-3}, and the sources at large wavenumbers and late times were wrong by more than their size.	The test that the copy reproduces the main project at zero rate.	The closure was removed from the Rust program. It is treated analytically (RS5, Eq. (3.5): inside the horizon it is the closure “Pressure”) and integrated for pressureless matter with an implicit method in the independent script.	RS5, Sec. 3.2; check `Idef-P6`
R-E37	A second design computed the derivatives of the metric potentials that the source of the temperature needs by numerical differences along the Runge–Kutta solution. The background of the program is a table with linear interpolation; second derivatives through it are meaningless, and the source had the wrong sign at one of the tested times.	A test against the analytic expression at a negligible rate.	The gaps are computed from the two identities of RS5, Sec. 3.1, as algebraic expressions; only the derivative of the (small) shear gap is taken by a difference, of first order and along the solution.	`eb_boltzmann.rs`; unit tests of `spectra_defl.rs`; check `def-E5`
R-E39	Two tests of the new program were themselves wrong. (i) The test at a “negligible” rate used 10^{-12}, at which the adaptive integrator chooses other steps and the spectra differ at 10^{-9}; the rate 10^{-30} takes the same branches and gives identical bits. (ii) The gap of the equation of the shear was first compared with a_{sp}gsum times a potential, which has the wrong dimension; the measure is $(a_{\mathrm{sp}}\mathrm{gsum}/H_{\mathrm{cf}})$ times the terms of the equation.	The tests failed, or gave numbers that changed with the unit of length.	Corrected as stated.	unit tests of `spectra_defl.rs`; check `def-E5`

3. Errors of notation (Hard Rule 5)

No.	What was wrong	How it was found	Correction	Where to check
R-E4	The author’s names `g55`, `g66`, `g77` for the rates were replaced by a Greek letter with an index, on the ground that they resemble the components of the metric.	The author objected.	The author’s names are used exactly; the components of the metric are distinguished by their subscripts and by the rule of RP1, Sec. 2.	RP1; check RS4

No.	What was wrong	How it was found	Correction	Where to check
R-E16	Names that the main registry uses for other objects were used for new objects in data files and statements (among them `tau`, `omega_c`, `mu`, `nu`, `kappa`, `eps`, `Q`, `M`, `G`, `D`, `f`, `u`, `A`, `P`). During the renaming, a first rule would also have changed the identifier of a check; it was restricted before it was applied.	Systematic comparison with the main registry and with the symbols of the notebooks.	All renamed; six machine checks now guard the registry.	RP1
R-E19	The first draft of DMD used unregistered letters (ε, ν, N, C, z, Y, ξ, E_0) and the program name `h0`, which is a symbol of a notebook; and it called the affine parameter the proper time "multiplied by" the mass (it is divided by the mass).	Re-reading against the registries; check RS3.	Replaced by registered symbols before the first commit of the document.	DMD; RP1
R-E20	The renaming of R-E16 replaced the gap matrix `M` by `m_bos`, the name of a boson mass, in two statements.	Reading the output while writing RS1.	Restored as `M_gap`.	check `def-K3`
R-E22	The scalar mode and its exponents were called `S` and `sigma` in headers of data files and in a figure; both letters have other meanings in the main registry.	The same reading.	`phi_bos` and `exponent`.	`data/scalar_zero_modes.csv`, `data/static_source.csv`
R-E31	In the draft of the manuscript on the 4+4 spacetime the matrix of the conserved charge was written W, while the same paper writes the kernel of the gap equation $W(k,k')$; the main registry calls the matrix W_{1644}. The matrix C^- was used there one section before it was defined. The draft of the manuscript on the 7+1 spacetime distinguished the sectional curvature K_{AB} from "the matrix K", which only the other paper has. And the tables of symbols of both drafts omitted symbols that the text uses (among them χ, β, α^j, K_{AB}, w, $Z_\pm$, ρ, ω_b, ω_c, v_F, ϕ, $W(k,k')$).	A script listed every letter that occurs in a formula of a manuscript and not in its table of symbols; each was then read in its place.	W_{1644} everywhere; C^- defined at its first use; the remark restricted to the paper that needs it; the tables completed; the sectional curvature and the index letters entered in the registry.	manuscripts, Sec. III and Appendix A; RP1, Sec. 3
R-E32	The manuscripts label the four flat coordinates other than the time by the index j and keep a for the four extra axes; DMD and the statements of the checks label the flat coordinates by a, as the corresponding documents of the main project do. The registry of the revision did not record the index letters at all, and the formula of the program for the Einstein and Lanczos tensors (curvatures on pairs of directions) was named in DMD but not written out.	Comparison of the registry with the manuscripts when the sections of the manuscripts were entered in it.	The registry lists the index letters with their ranges, RP1 explains the rule, and DMD writes out the formula with the sectional curvatures K_{AB}.	RP1, Sec. 3; DMD, Sec. 1.4a

No.	What was wrong	How it was found	Correction	Where to check
R-E40	The new data files of the spectra used the column name `tau` both for the optical depth and for the conformal time; the bare name is retired in the revision (R-E16), and the main registry gives τ to the optical depth. The registry entry of the metric perturbations listed the program name `hprime`, which is a symbol of one of the author's notebooks. The letters h, η, α, θ, σ of the standard reference for the perturbation equations all have other meanings in the project.	The checks RS3 and RS5 of the symbols failed; the letters were compared with the registries before RS5 was written.	Columns `tau_reio` and `tau_cf`; the program name removed from the registry; the document symbols h_{sy}, η_{sy}, α_{sy}, τ_{cf} and script letters for the sums over matter (registry group D9).	RP1; RS5, Sec. 0.2

4. Errors of procedure

No.	What was wrong	How it was found	Correction	Where to check
R-E15	A figure asked for matter of the Fermi momentum $2\,m$, for which no history had been computed.	The figure script failed.	The figure shows the three densities that exist.	`figures/fig_matter_history`
R-E24	A commit (`d5ddbc7`) was made without running the unit tests. One test asserts the number of exact geometry checks, which had grown from 14 to 16, and failed. The checks themselves all passed.	The tests were run for the provenance RP8.	The test was corrected in the next commit; the tests are run before every commit.	`cargo test`
R-E27	The command `matter` contained three numbers copied by hand from the output of the command `cosmology`: the bounds 0.0644 and 0.234 (rounded), and $H_0 t_0 = 0.9507$ for the age of the model, whereas the model of the revision has $H_0 t_0 = 0.9516$. The last digits of the table of drifts (RS3, Sec. 6.3) and one digit of a growth factor (1.0590 instead of 1.0589) were affected; no conclusion was.	Reading the program while tracing a number for the check of the numbers.	Both commands now take the bounds and the age from one routine (`rate_bounds`); the tables were regenerated and the two sentences that quoted the old digits corrected.	`cosmo_defl.rs`; checks `def-D4`, `def-D5`; `data/dm_drift.csv`
R-E29	The table of source files of RP8 gave the numbers of lines, typed by hand; after later changes of the program two of them were out of date.	The same check, which found a number that no program states.	The table, the number of files, of lines, of unit tests and of figures are now written by the script that writes the other tables.	RP8, Sec. 3

No.	What was wrong	How it was found	Correction	Where to check
R-E41	Four checks of the new commands had criteria that were written before the result was known and were not consequences of anything: that the Hubble constant falls by less than 0.2 km/s/Mpc at the bound of lunar laser ranging (it falls by 0.211); that the fit at the rate of nucleosynthesis has $\Delta\chi^2 > 25$ for Planck-like data (it has 11.6); that the fits of the dark-radiation families on the deflating background differ from those of the main project by less than 1 in $\Delta\chi^2$ (they are systematically lower, by more); and the statement of `def-D4` said that the sign of a change is not a prediction where both closures give the same sign.	The checks failed, or the statement contradicted its own numbers.	The criteria test consistency (the fit against the Fisher matrix, one closure against the other, the copy against the main project); the statements report the numbers. The third case led to the joint analysis of the rate and the extra radiation (check `def-N4`).	checks `def-C3`, `def-F4`, `def-N2`, `def-N4`, `def-D4`
R-E42	The test option `quick` of the command `families` wrote its incomplete tables over the data files of the full run.	Noticed when the instructions of RP8 were written.	The test writes to a temporary folder.	`cmd_spectra.rs`; RP8, Sec. 2

5. What the list shows

Forty-three errors in one day of work is not a small number, and the reader should draw the right conclusion from it: the statements of the revision are to be trusted to the extent that a machine check stands behind them, and no further. Of the errors of physics, most were found because a check with a definite expectation failed; two (R-E7, R-E23) were found only by re-reading, and concerned the *interpretation* of a correct calculation. Two more (R-E33, R-E34) had passed every check, because the checks tested the program against the equation that had been written down and not the equation against the theory; they were found only when the calculation was done again from the eight-dimensional equations (R-E33) and with a more complete program (R-E34). Statements of interpretation in the documents of the revision — in particular the verdicts of RS2 and RS3 — deserve the reader's independent scrutiny more than the numbers do.

The entries R-E25 to R-E29 were found while the two manuscripts were written, when every number quoted in the running text was compared with what the programs state. None of them changes a conclusion, but each was a number that was typed or copied by hand where a program should have supplied it. The comparison is now a script (`code/scripts/check_numbers.py`; RP8, Sec. 4), which is run before a document is released. The last three entries (R-E30 to R-E32) were found when the finished manuscripts were read once more, page by page, and when their symbols were compared with their tables of symbols and with the registry. They concern a unit and the notation, not a result.

The entries R-E33 to R-E43 belong to the Einstein–Boltzmann calculations of the supplement RS5. Three of them (R-E33, R-E34, R-E35) corrected statements that earlier versions of the documents of the revision had made; the corrections are marked in those documents where they occur.